

Avatars for Good Drinking: An Exploratory Study of The Effects of Avatar's Body Shape on Beverage Perception

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ABSTRACT

The Proteus effect is known as the effect that the self-image evoked by an avatar influences the behavior. Previous studies have shown that the Proteus effect influences food-related behaviors, but it is not clear whether it influences perceptions of food. Therefore, in this study, we investigated whether the avatar's body shape affects beverage perception when drinking a beverage in a virtual environment. Specifically, we measured the sense of body ownership toward the avatar, the taste of the beverage, the amount of the beverage consumed, and the purchase intention of the beverage. The results showed that a gradual transition in body shape with drinking produced a significant improvement in the sense of body ownership, particularly in terms of feeling that the virtual body belonged to oneself. We also report the larger body shape significantly increases the purchase intention. Furthermore, we found that image congruence between the avatar and cola induces better taste perception.

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1 INTRODUCTION

The number of users of social virtual reality (VR) platforms continues to increase. Among them, eating and drinking using VR is a unique activity. The Proteus effect is known as the effect that the self-image evoked by an embodied avatar, which is a virtual body in VR influences the behavior of users [Yee and Bailenson 2007]. Previous studies have not only confirmed that the perception of one's body shape changes with the shape and size of the avatar but have also shown that the virtual body shape can induce food-related stereotypes, such as a greater tendency to avoid high-calorie foods when a slim avatar is used, thereby unconsciously modulating behavior [Tambone et al. 2021]. However, it needs to be clarified whether the body shape of the avatar influences perceptions of



Figure 1: Avatars used in the experiment (child / standard / muscular)

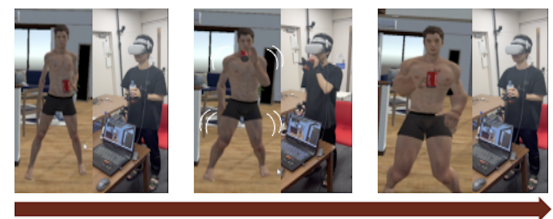


Figure 2: Body shape transition depending on the amount of the beverages consumed.

food. Therefore, we first explored whether the avatar's body shape affects beverage perception when drinking a beverage in a virtual environment. Specifically, we measured the sense of body ownership toward the avatar, the beverage's taste, the amount of the beverage consumed, and the purchase intention of the beverage with different size of avatar.

Furthermore, based on previous research suggesting that a smooth spatio-temporal transition from real to virtual environments maintains the "conviction about reality" [Okeda et al. 2018], we also examined whether a gradual transition from standard avatars, which is expected to have the highest sense of body ownership due to its proximity to the real body of a typical participant, to muscular avatars and then to child avatars would amplify the effect.

2 EXPERIMENT

Because only male avatars of different body shapes were available, the participants in this experiment were limited to males. Twelve males (aged 22.91 ± 1.31) participated in the study. They wore an Oculus Quest2 and HTC VIVE Trackers on their hands and used their assigned avatars according to the avatar condition. Five avatar conditions were tested: three avatar body types a) standard, b) muscular, and c) child and two conditions in which the avatar changes

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from d) standard to muscular or e) standard to child depending on the amount of the cola consumed. The experimental procedure is as follows: First, to ensure the sense of body ownership, participants moved their avatar in front of a mirror for 2 minutes and walked in a virtual environment using a controller for 2 minutes. After that, participants drank cola at their own pace for 3 minutes, and rated their *sense of body ownership* (Virtual Embodiment Questionnaire; VEQ), *perceived flavor* and *purchase intention* toward cola (7-point Likert scale for each). Participants in conditions d) and e) responded to the VEQ regarding the virtual body after its shape had changed. We also measured *the amount consumed* of cola for each trial.

The hypotheses of this experiment is twofold: (1) Larger body size correlates with more *amount consumed*, weaker *perceived flavor*, and higher *purchase intention*. (2) By making participants feel that drinking beverages has changed their body shape, the body ownership can be maintained even when the physical characteristics of the avatar differ significantly from their real body.

3 RESULT

A steel test showed no significant differences between conditions using muscular or child avatars and the control condition with standard avatars.

Moreover, aligned rank transform was applied to apply ANOVA to each nonparametric data. Then a two-way ANOVA was conducted to examine the effects of avatar body shape (muscular and child), transitions (with-transition and without-transition), and their interaction effect. As a result, a significant main effect of transitions on the sense of body ownership was observed in one of its metrics, specifically the questionnaire item “*It felt like the virtual body belonged to me.*” ($F(1,8) = 5.171, p = 0.0278, \eta_p^2 = 0.105$) (Fig.3 left). However, no significant difference was observed in the scores of Body Ownership scale. Additionally, a significant main effect of avatar’s body shape was found for the same questionnaire items ($F(1,8) = 6.583, p = 0.0137, \eta_p^2 = 0.130$) (Fig.3 left). The test also revealed a significant main effect of avatar’s body shape on the scores of Body Ownership ($F(1,8) = 4.683, p = 0.0359, \eta_p^2 = 0.096$). Several participants reported that moving naturally to avoid obstacles when their avatar changed from standard to muscular with drink consumption, but not when a muscular avatar was used from the start. The test also showed a significant effect of body shape on purchase intention ($F(1,8) = 4.554, p = 0.0384, \eta_p^2 = 0.093$) (Fig.3 center).

Regarding food perception, no significant effects of avatar shapes or transitions were observed. Meanwhile, there was a notable correlation between participants’ image congruence with cola and avatars and the perceived strength of the aftertaste ($r = 0.4772$), drinkability ($r = 0.4509$), preference for the cola used in the experiment ($r = 0.3774$), and purchase intention for the cola ($r = 0.6058$). As for the amount consumed, no noticeable trend was observed from the quantitative data (Fig.3 right). However, participants’ feedback indicated that cola was perceived as the most drinkable when the standard avatar was used among the three avatars.

4 DISCUSSION

The results supported our hypothesis(1) on purchase intention but not the amount of beverage consumed or food perception. On the other hand, image congruence between avatar and cola correlates

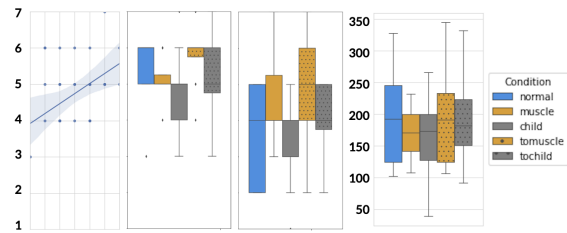


Figure 3: Taste Perception(Correlation between image congruence with cola and avatar(x-axis) and the perceived strength of aftertaste(y-axis))/ Body ownership / Purchase intention / The amount of beverage consumed

with better taste. These results suggest that using avatars with a specific appearance that matches the image of a particular beverage substantially impacts changes in food perception. Hypothesis(2) was partially confirmed, as we found that we could maintain a sense of body ownership by gradually transitioning the size of avatar according to consumption. Moreover, post-experiment feedback from participants suggested that some participants experienced an even more profound sense of body ownership than when using the standard avatar. This finding suggests that transitions may be utilized to enhance a sense of body ownership in humanoid avatars or even non-human entities or objects with characteristics that differ significantly from those of the real body. However, it is still unclear whether the gradual transition is essential in improving body ownership or whether providing causal attribution in food and drink intake plays a significant role.

5 CONCLUSION

We conducted a preliminary study of the effects of avatar shape and spatio-temporal transitions on the sense of body ownership, food perceptions, food consumption, and purchase intention. We suggest that (1) Body shape transitions based on the amount of the beverage consumed can potentially improve the sense of body ownership, (2) Body size tends to influence purchasing intention, and (3) Image congruence between the avatar and cola induces better taste. However, since this study only included a limited number of people and gender, the results cannot be overly generalized.

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